

Business Model Design and the Performance of Entrepreneurial Firms

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We focus on the design of an organization's set of boundary-spanning transactions—business model design—and ask how business model design affects the performance of entrepreneurial firms. By extending and integrating theoretical perspectives that inform the study of boundary-spanning organization design, we propose hypotheses about the impact of efficiency-centered and novelty-centered business model design on the performance of entrepreneurial firms. To test these hypotheses, we developed and analyzed a unique data set of 190 entrepreneurial firms that were publicly listed on U.S. and European stock exchanges. The empirical results show that novelty-centered business model design matters to the performance of entrepreneurial firms. Our analysis also shows that this positive relationship is remarkably stable across time, even under varying environmental regimes. Additionally, we find indications of potential diseconomies of scope in design; that is, entrepreneurs' attempts to incorporate both efficiency- and novelty-centered design elements into their business models may be counterproductive.

Key words: organization design; new organizational forms; business model; design themes; organization performance; environmental munificence

Substantial research on organization design has focused on internal design issues such as centralization, span of control, personnel ratios, and lines of authority (e.g., Nystrom and Starbuck 1981). Some scholars, however, have observed that organizations are increasingly “experimenting with their governance of transactions, that is, adopting new ways of structuring their boundaries” (Foss 2002, p. 1). A growing body of work on organizational forms has gradually shifted attention from internal design toward modes of organizing and managing transactions with the firm's environment (e.g., Ilinitch et al. 1996, Lewin and Volbverda 1999, Miles and Snow 1986, Romanelli 1991). While this body of research has enhanced our understanding of how managers and entrepreneurs set organizational boundaries, important questions remain. For example, how can the design of an organization's set of boundary-spanning transactions be described and measured? And what do we know about the performance implications of different designs?

Recent advances in communication and information technologies, such as the emergence and the swift expansion of the Internet, and the rapid decline in computing and communication costs, have accentuated the possibilities for the design of new boundary-spanning organizational forms (Daft and Lewin 1993, Dunbar and Starbuck 2006, Foss 2002, Ilinitch et al. 1996). Indeed, these developments have opened new horizons for the design of business models by enabling firms to

fundamentally change the way they organize and engage in economic exchanges, both within and across firm and industry boundaries (Mendelson 2000). According to Brynjolfsson and Hitt (2004), this includes the ways in which firms interact with suppliers as well as with customers. An emerging stream of work on boundary-spanning designs complements a large body of literature that points to the links between internal organization design issues, such as the degree of decentralization, the structure of incentives, and the implications on productivity for investment in information technologies (e.g., see Bresnahan et al. 2002 or Ichniowski et al. 1996). This paper builds on the shift in perspective from viewing organization form as a complement to IT investments toward viewing IT as an enabler of boundary-spanning organizational design.

In this paper we refer to the design of an organization's boundary-spanning transactions as business model design, and we ask how business model design can be measured, and how it affects firm performance. We formally define the business model as depicting “the content, structure, and governance of transactions designed so as to create value through the exploitation of business opportunities” (Amit and Zott 2001, p. 511). A business model elucidates how an organization is linked to external stakeholders, and how it engages in economic exchanges with them to create value for all exchange partners.

Designing the business model is a salient issue for entrepreneurial firms who are less constrained by path dependencies and inertia than more established firms (Stinchcombe 1965). Following Bhidé (2000), we define entrepreneurial firms as relatively young organizations that have the potential of attaining significant size and profitability. These firms must solve coordination problems in a world of novelty and systemic change (Langlois 2005). Their performance is therefore often critically dependent on boundary-spanning organizational arrangements (Hite and Hesterly 2001). One of the central design tasks of entrepreneurs is to delineate the ways in which their new businesses transact with suppliers, customers, and partners. As Ireland et al. (2001, p. 53) note, entrepreneurs often “try to find fundamentally new ways of doing business that will disrupt an industry’s existing competitive rules, leading to the development of new business models.” For example, Christensen (2001) highlighted the shift in the locus of profitability in the computer industry as companies like Dell pioneered nonintegrated and flexible business models in which production and distribution were organized in novel ways. Even when entrepreneurial firms replicate the business models of existing organizations (Aldrich 1999), they may have to adapt these designs to their own particular market niche (McGrath and MacMillan 2000).

Although recent work in entrepreneurship and organization theory has begun to address the important role of design in the entrepreneurship process (Hargadorn and Douglas 2001, Romme 2003, Van de Ven et al. 1984), relatively little is known about the specific trade-offs and performance implications of business model design, which can be far-reaching. For example, Hargadorn and Douglas (2001, p. 494) attribute the failure of Prodigy, an online service in which investors had invested \$600 million, to the mismatch between the design of its business model and customer needs.

In this paper we identify two critical themes of business model design—efficiency centered and novelty centered—and offer hypotheses about the impact of business model design themes on the performance of the focal entrepreneurial firm, taking into consideration the potentially moderating role of the environment. Efficiency-centered business model design refers to the measures that firms may take to achieve transaction efficiency through their business models. It aims at reducing transaction costs for *all* transaction participants. Novelty-centered business model design refers to new ways of conducting economic exchanges among various participants. It can be achieved, for example, by connecting previously unconnected parties, by linking transaction participants in new ways, or by designing new transaction mechanisms. Efficiency- and novelty-centered designs are neither orthogonal (for instance, novel design elements may engender lower transaction costs), nor are they mutually exclusive: Both may be

present in any given business model. They are not exhaustive, either. Business models may be characterized by other value-creation themes. These could include “lock-in” designs, which attempt to retain stakeholders, and “complementarities” designs, which emphasize the bundling of goods, activities, resources, or technologies (Amit and Zott 2001). However, here we focus on efficiency- and novelty-centered designs in the interest of building and testing a parsimonious theory.

To test our hypotheses, we have developed a unique data set that contains detailed information about the business models of 190 entrepreneurial firms that were listed on U.S. or European public exchanges between 1996 and 2000. We measure each business model design theme as a variable at a particular point in time, and we regress these variables on a range of performance measures. Overall, we find that business model design matters to the performance of entrepreneurial firms. Our most robust finding relates to the positive association between novelty-centered business model design and firm performance. Our analysis shows that this positive relationship is remarkably stable across time, even under varying environmental regimes. Our results also indicate that entrepreneurs’ attempts to design both efficiency- and novelty-centered business models may be counterproductive.

This paper builds on and extends earlier work that has considered business models in the context of organizational performance. Focusing primarily on the impact of network effects on the stock market value of e-commerce firms, Rajgopal et al. (2003) examine how network effects interact with the firm’s business model, measured as a categorical variable (i.e., content provider, portal, financial services, e-tailer, or auction site). In our analysis, we focus primarily on the design of the business model (rather than on network effects), and on its direct impact on firm performance, theoretically as well as empirically. Our proposed measures of business model design are continuous and apply to a broad range of firms. Filson (2004) studies the impact of the competitive strategies of Amazon.com, Barnesandnoble.com, CDNow, and N2K on firm value. We, too, focus on firms that derive at least some of their revenue through transactions that are executed on the Internet. However, we examine in detail the impact on firm value of business model design, distinct from the competitive strategy of a firm (Zott and Amit 2006). Our sample size is also large.

Our study attempts to make several contributions to the organization design literature. First, we refine concepts and measures for examining the design of a firm’s business model. The importance of research on transaction designs as new organizational forms has been recognized in earlier studies, (e.g., Foss 2002, Rindova and Kotha 2001). We contribute to this literature through the

development of fine-grained concepts for operationalizing and measuring business model designs. Second, we provide a theoretical extension of the transaction costs perspective and Schumpeter's theory of innovation. By integrating these theories with bargaining theory, we develop the performance implications of business model design, specifically for entrepreneurial firms whose transactions are enabled by information and communication technologies. To date, little about this has been published. Third, drawing on a large and a unique data set about business model design themes, which we have collected ourselves, we test the links between these design themes and focal firm performance. Although we build on earlier studies that examine how business models are linked to firm performance (e.g., Rajgopal et al. 2003), we believe that this is the first study to operationalize, measure, and test the performance consequences of business model design themes. Our study also contributes to the literature on entrepreneurship, by highlighting the pivotal role that business model design plays in the performance of entrepreneurial firms.

To summarize, in this paper we argue theoretically and show empirically that the business model is a useful unit of analysis for research on boundary-spanning organization design, as well as a locus of innovation that has so far been largely overlooked by entrepreneurship research. Rindova and Kotha (2001, p. 1277) have pointed out the need for "a broader and more dynamic understanding of [organizational] form, in which it is viewed as a flexible arrangement of resources and structures configured to generate a stream of value-creating products and services." The concept of the business model fulfills these requirements, and can potentially help advance the emerging body of research on new organizational forms.

The next section presents our theory and hypotheses and is followed by sections describing our data, methods, and our results. We conclude with a discussion of our findings and the implications of our study for future research.

Theory and Hypotheses Development

Business Model Design Themes

Configuration theory provides a useful starting point for developing measures of business model design, because it considers holistic configurations, or *gestalts*, of design elements (Miles and Snow 1978). Configurations are constellations of design elements that commonly occur together because their interdependence makes them fall into patterns (Meyer et al. 1993). The design elements of a business model are the content, structure, and governance of transactions. In this paper, we follow Miller's (1996) suggestion to view configuration as a variable rather than as a deviation from an ideal type. Miller states that, "Configuration...can be defined as

the degree to which an organization's elements are orchestrated and connected by a single theme" (Miller 1996, p. 509).

So what are the common design themes that orchestrate and connect the elements of a *business model*? Miller (1996) identifies innovation and efficiency. His choice is particularly appropriate for the study of business models adopted by entrepreneurial firms because innovation and efficiency reflect fundamental alternatives for entrepreneurs to create value under uncertainty. Entrepreneurs can create new designs and/or reproduce and copy existing ones (Aldrich 1999). Imitation-based approaches toward business creation are often associated with an emphasis on lower costs, i.e., increased efficiency (Zott 2003). Because these themes are not mutually exclusive, any given business model design can be novelty centered and efficiency centered at the same time.

Business Model Design, Firm Performance, and the Moderating Role of the Environment

We hypothesize that the design of an entrepreneurial firm's business model, which is centered specifically on the themes of novelty and/or efficiency, is associated with the firm's performance. This association can be broken down into two effects: One relies on the total value-creation potential of the business model design, and the other considers the impact of business model design on the firm's ability to appropriate the value that its business model creates.

Business models can create value either by enhancing the customers' willingness to pay or by decreasing suppliers' and partners' opportunity costs—for example, through improved transaction efficiency. The total value created by a business model is also a function of the competitive alternatives, in other words, the market power of the focal firm's business model vis-à-vis rival business models. The total value created is the value created for all business model stakeholders (focal firm, customers, suppliers, and other exchange partners). It is the upper limit for the value that can be captured by the focal firm (Brandenburger and Stuart 1996).

How does business model design influence the competing claims to total value created by different stakeholders in the business model? Drawing on Brandenburger and Nalebuff (1995) and Brandenburger and Stuart (1996), we reason that there is a positive association between the design of the business model and the performance of the focal firm—if, for a given level of competition, the focal firm's business model design creates value, *and* it does *not decrease* its bargaining power relative to other business model stakeholders.

Environmental conditions seem important as moderators of the hypothesized relationship between business model design and the performance of an entrepreneurial firm (McArthur and Nystrom 1991). Munificence, dynamism, and complexity are all important dimensions

of the environment (Dess and Beard 1984). We focus on munificence, which captures the availability of resources, and which—according to Randolph and Dess (1984) and McArthur and Nystrom (1991)—influences the survival and growth of existing companies as well as the ability of new firms to enter the market. The use of munificence as a focal dimension of environmental uncertainty derives from a resource-dependence view of organizations, which treats environments as arenas in which all compete for resources (Pfeffer and Salancik 1978, Aldrich 1979). This perspective seems particularly relevant for entrepreneurial firms, which depend on external resources, and on the receptivity of capital and product markets to innovative activity.

Building on the work of Aldrich (1979), Dess and Beard (1984), and Tushman and Anderson (1986), we define munificence as the extent to which the environment supports growth. In this paper, munificence refers to the scarcity or abundance of critical resources required to create and implement business model designs. The performance prospects of ventures with novelty- and/or efficiency-centered business model designs may vary with the availability and costs of resources that entrepreneurs can access.

Novelty-Centered Business Model Design and Performance

The essence of novelty-centered business model design is the conceptualization and adoption of new ways of conducting economic exchanges, which can be achieved, for example, by connecting previously unconnected parties, by linking transaction participants in new ways, or by designing new transaction mechanisms. Business model innovation may complement innovation in products and services, methods of production, distribution or marketing, and markets (Schumpeter 1934). A novel business model either creates a new market (like eBay) or innovates transactions in existing markets (like Priceline.com). Dell, for example, implemented a customer-driven build-to-order business model that replaced the traditional build-to-stock model of selling computers through retail stores (Brynjolfsson and Hitt 2004).

Not only can the business model exploit an opportunity for wealth creation, but its design may in itself be part of the opportunity development process. The entrepreneur-as-designer can cocreate opportunities, for example, by drawing on new information and communication technologies to bridge factor and product markets in new ways.

Business model innovation may give rise to entrepreneurial rents (Rumelt 1987). These monopoly-type rents may accrue to business model stakeholders between the introduction of an innovation and its diffusion. Although we expect novelty-centered business model design to have a positive primary effect on the performance of entrepreneurial firms, entrepreneurial

rents may accrue to *all* stakeholders in the business model. To predict the overall effect of novelty-centered business model design on the performance of the focal entrepreneurial firm, we must also consider its effect on the firm's ability to appropriate the value that its business model generates. This ability depends on factors like (i) the switching costs of other business model stakeholders, (ii) the focal firm's ability to control information, (iii) the ability of other stakeholders to take unified action vis-à-vis the focal firm, and (iv) the replacement costs of other stakeholders (see Coff 1999, p. 122). For example, the higher the switching costs of other business model stakeholders, the greater the focal firm's bargaining power vis-à-vis these stakeholders, and the greater its ability to appropriate rent.

We suggest that, on average, an increase in business model novelty will not decrease the focal entrepreneurial firm's ex post bargaining power relative to other business model stakeholders. The focal firm is the innovator, and its business model is the locus of innovation. The higher the degree of business model novelty, the higher the switching costs for the focal firm's customers, suppliers, and partners, as there may not be readily available alternatives to doing business with the focal firm. The other determinants of the focal firm's bargaining power, identified by Coff (1999), are unlikely to be systematically affected in one direction or the other. The focal firm's bargaining power vis-à-vis these parties is unlikely to be diminished through novelty-centered business model design. Therefore, considering the positive effect of novelty-centered business model design, both on total value created and on the ability of the focal firm to capture that value, we expect a positive effect of novelty-centered business model design on the performance of an entrepreneurial firm.

HYPOTHESIS 1. *The more novelty centered an entrepreneurial firm's business model design, the higher the firm's performance.*

Novelty-centered business model design will matter more to performance in periods of high resource availability, when entrepreneurs' dynamic governance costs (e.g., persuasion costs) are lower (Langlois 1992, Langlois and Robertson 1995), than in periods of resource scarcity. They can persuade, negotiate, and coordinate with resource holders more easily, and teach them about the merits of their innovative business model designs. They have easier access to the resources needed to support and implement their business model innovations, such as investments in complementary assets. Entrepreneurial firms with novelty-centered business model designs are poised to take advantage of the greater willingness of customers to spend, and they will not suffer a decrease in their aggregate bargaining power vis-à-vis other business model stakeholders. Thus, novel business model design is more distinctly associated with

higher firm performance when resources are abundant than when they are scarce.

HYPOTHESIS 2. *In environments characterized by high resource munificence, the positive association between novelty-centered design and the performance of the entrepreneurial firm will be stronger than in environments with low resource munificence.*

Efficiency-Centered Business Model Design and Performance

An alternative way for entrepreneurs to create wealth is to imitate rather than innovate—to do things similar to established organizations, but in a more efficient way (Aldrich 1999, Zott 2003). To examine the performance implications of efficiency-centered business models, we build on the transaction cost perspective (Milgrom and Roberts 1992; Williamson 1975, 1983), which refers to the design of economic transactions. According to Williamson (1983), exchange attributes, including information asymmetry and complexity, determine that transactions will be organized into markets or hierarchies in ways that minimize transaction costs and maximize performance. Researchers generally assume that economic actors whose transactions are unaligned with appropriate governance structures are “more likely to display poor financial performance...than those whose transactions are properly aligned” (Silverman 2001, p. 484). Poppo and Zenger (1998) have explicitly modeled the performance implications of the transaction cost perspective, and Milgrom and Roberts (1992) have elaborated on the effect that transaction costs, in the form of coordination, and motivation costs have on firm performance. These studies suggest that there is an important direct relationship between the design of transactions and firm performance.

Efficiency-centered design refers to the measures that firms may take to achieve transaction efficiency through their business models. The construct focuses on business model design and is not intended to capture all means by which a firm can strive for efficiency (e.g., through a reduction of production costs). The essence of an efficiency-centered business model is the reduction of transaction costs. This reduction can derive from the attenuation of uncertainty, complexity, or information asymmetry (Williamson 1975), as well as from reduced coordination costs and transaction risk (Clemons and Row 1992, Langlois 1992, Milgrom and Roberts 1992). The order-tracking feature in Amazon’s business model, for example, is aimed at enhancing transaction transparency and therefore constitutes an efficiency-centered design element. It reduces the cost of providing information to the logistics company, and induces more customers to check on their packages (Brynjolfsson and Hitt 2004). Other efficiency-centered design elements are intended to increase the reliability and simplicity

of transactions, reduce the asymmetry of information among transaction participants, speed up transactions, enable demand aggregation, reduce inventory, provide for transaction scalability, or reduce the direct and indirect costs of transactions. Consider Baxter ASAP, which lets hospitals order supplies electronically direct from wholesalers. By reallocating its saved resources (the costs of data entry), the company was able to offer additional value-adding services to its customers (Brynjolfsson and Hitt 2004). Consequently, we expect a positive primary effect on firm performance of adopting an efficiency-centered business model design.

As with the novelty-centered business model design, to predict the overall effect of an efficiency-centered design on firm performance, we must consider its effect on the firm’s ability to appropriate the value that it generates. Efficiency-centered business model design aims at reducing transaction costs, for example, through simplified transactions, reduced transaction complexity, or deep linkages among business model stakeholders that often do not require transaction-specific investments (web services, for example). These characteristics of efficiency-centered business model design are likely to affect the switching costs for *all* business model stakeholders in the same direction so that, in the aggregate, the balance of power among these parties will not shift. Importantly, the focal firm’s bargaining power is unlikely to decrease.

Another central aspect of efficiency-centered design is that it enables better information flow among stakeholders and reduces information asymmetries among the parties, limiting the control over information that any stakeholder can have. In general, this aspect does not negatively affect the focal firm’s bargaining power. Moreover, the third determinant of the focal firm’s bargaining power identified by Coff (1999, see above)—the ability of other business model stakeholders to take unified action against the focal firm—is unlikely to be systematically affected in one direction or the other by design efficiency. Last, it should be noted that reducing direct transaction costs (e.g., search, transportation, and coordination costs) increases the pool of potential customers, as well as partners and suppliers, and implies a reduction in the cost to the focal firm of replacing such stakeholders, increasing the firm’s bargaining power.

These arguments suggest that, on balance, a more pronounced efficiency-centered business model design does not decrease the focal firm’s bargaining power relative to other business model stakeholders. We therefore expect a positive main effect of efficiency-centered business model design on the performance of an entrepreneurial firm.

HYPOTHESIS 3. *The more efficiency centered an entrepreneurial firm’s business model design, the higher the firm’s performance.*

When resources are scarce, efficiency-centered business model design assumes greater importance as a differentiating factor than in periods of resource munificence. In tough economic environments, consumers and businesses spend and invest less, and cost savings become more important as a driver of value creation. Entrepreneurial firms are more volatile than established organizations (Stinchcombe 1965), so are sensitive to such changes. Conversely, during times of high environmental munificence, total value can be enhanced, for example, by tapping additional revenue streams. In other words, in environments characterized by low resource munificence, the advantages derived from reduced transaction costs are accentuated, while our arguments about the bargaining power of firms with efficient business models continue to hold. Thus, efficient business model design will be more distinctly associated with higher performance of an entrepreneurial firm when resources are scarce than when they are abundant.

HYPOTHESIS 4. *In environments characterized by low resource munificence, the positive association between efficiency-centered design and the performance of the entrepreneurial firm will be stronger than in environments with high resource munificence.*

Interaction Between Novelty- and Efficiency-Centered Business Model Design and Performance

Do these arguments imply that entrepreneurs should embrace both efficiency-centered and novelty-centered business model designs? The need to balance design elements has been recognized by researchers who highlight the benefits to entrepreneurs of reconciling distinct aspects of design, such as the familiar and the unfamiliar (Hargadorn and Douglas 2001), conformity and differentiation (Deephouse 1999), and reliability and distinctiveness (Lounsbury and Glynn 2001). Achieving this balance can help entrepreneurs build much-needed legitimacy (Zott and Huy 2006), a prerequisite for venture growth and performance (Zimmerman and Zeitz 2002). This suggests that novelty and efficiency can be complementary, and that their interaction could have a positive effect on performance. First, increasing the degree of novelty of a business model may enhance the return on efficiency-centered design. Novelty-centered business model design makes a business model more distinctive, and this may result in increased switching costs for other business model stakeholders because of fewer comparable alternatives. By emphasizing business model novelty, the focal firm may be better positioned to appropriate some of the value it creates through increased efficiency. Second, increasing the emphasis on efficiency-centered design may enhance the return on design novelty. Novel business models that are also designed for efficiency may appeal to a wider range of customers (i.e., not only to those who are intrigued by its novel elements, but

also to those who appreciate lower transaction and coordination costs). Thus, by simultaneously emphasizing efficiency and novelty as design themes, the entrepreneur may be able to create even more value than through either novelty-centered or efficiency-centered business model design alone.

HYPOTHESIS 5. *The more novelty centered and efficiency centered the business model design, the higher the performance of the entrepreneurial firm.*

However, another line of reasoning suggests that attempts by entrepreneurs to design their business models for both higher efficiency and greater novelty may instead adversely affect their firm's performance. Embracing two major design themes concurrently could lead to suboptimal resource allocation. Given the limited resources available to entrepreneurial firms, entrepreneurs who try to achieve too much at once may find that they are not obtaining adequate returns on their design efforts and investments. This is because a lack of focus may confuse market participants, undermine the venture's legitimacy, create technological and organizational problems, and lead to higher costs. Furthermore, a firm that gets stuck between innovation and imitation—or, analogously, between novelty and efficiency design themes—may perform poorly because it misses out on the opportunity to learn to become an even more skillful innovator or imitator (Zott 2003). In summary, there might be diseconomies of scope in design resulting from bundling novel and efficient design elements.

HYPOTHESIS 6. *The more novelty centered and efficiency centered the business model design, the lower the performance of the entrepreneurial firm.*

Data and Methods

Sample

To test our hypotheses, we studied the business models of firms that derived all or part of their revenues from transactions conducted over the Internet, as these firms are likely to experiment with, and take advantage of, the possibilities that advanced information and communication technologies offer for the design of business models. We examined the business models of firms that went public in Europe or the United States between April 1996 and May 2000. Our sample selection strategy enabled us to create a data set of 362 relatively young entrepreneurial firms and their business models, from which we randomly sampled 201. Limited data availability forced us to drop 11 firms from the sample, which left us with a final sample size of 190. We considered public companies, both to ensure the availability of data and because data collection from initial public offering documents is an acknowledged method for studying entrepreneurial firms (e.g., Dowling and McGee 1994). Concerns about survival bias are mitigated by the

fact that during the sampling period, the threshold for becoming a public company was relatively low, which resulted in listing very young and immature companies. As a result, many of the firms in our sample still faced highly uncertain prospects at the time they went public (Lieberman 2005). Indeed, when we investigated the fate of these firms in June 2004, we found that 107 of our 190 sample firms (56%) had been delisted: 20 had gone bankrupt, 68 had been acquired, 8 had been merged, and 11 had been taken private.¹

Data Collection

For each business model design theme, we built composite scales and identified and measured relevant items in a survey instrument (see Appendix A). The surveying process proceeded in five stages: (1) development of the survey instrument, (2) development of measurement scales, (3) pretesting of the survey, (4) development of an online Web interface and of a central database, and (5) data collection.

Following the increasing use of panelists in management research (e.g., Iansiti and Clark 1994, Lee et al. 2003, MacCormack et al. 2001), we hired 11 part- or full-time research assistants (primarily MBA students), and trained them as raters to fill in the survey instrument for assigned sample companies. We carefully selected our raters from a larger pool of applicants by interviewing them and asking them to submit an abbreviated test survey on a randomly chosen sample company to display their understanding of Internet-based business models. After choosing the most qualified candidates, we trained them in data collection and data analysis. In addition, raters were provided with written guidelines on the proper way to address survey items. Each rater was assigned to one of two project managers, who reviewed completed surveys for internal consistency and completeness, but not for the accuracy of each individual measurement. On average, it took a rater about two and a half days to collect data on a given business model, to analyze the model, and to complete the survey. Data sources included IPO prospectuses, annual reports, investment analysts' reports, and websites. The data were collected from May 2000 to June 2001. During that time period, we took one measurement of the design themes for each of the 190 business models in our sample, collecting cross-sectional data on our independent variables.

We validated interrater reliability by assigning a randomly chosen business model to two different raters (each of whom was assigned to a different project manager), and by conducting a pairwise comparison of responses, yielding a Cronbach alpha of 0.81 and a Pearson correlation coefficient of 0.72. Raters were in broad agreement with each other for 82% of the individual items. We repeated the test periodically for different raters and different business models and found that all indicators of reliability had further improved.

Independent Variables

We selected two independent variables of business model design: design efficiency and design novelty. We used 13 items as measures of design efficiency, and 13 items as measures of design novelty. Given the difficulty of obtaining objective measures of business model design, we deemed the use of perceptual measures obtained from our raters appropriate (Dess and Robinson 1984). The strength of each of these items in a given business model was measured using Likert-type scales (see Appendix A for details) and coded into a standardized score. After coding, we aggregated the item scores for each design theme into an overall score for the composite scale using equal weights (see Mendelson 2000). This process yielded distinct quantitative measures of the extent to which each business model in the sample leveraged efficiency and novelty as design themes. (See Table 2 for summary statistics.)

We validated the internal consistency and reliability of our measures using standardized Cronbach alpha coefficients, 0.69 for the design efficiency measure and 0.72 for the design novelty measure. Our measures sufficiently satisfy Nunnally's (1978) guidelines, which suggest 0.7 as a benchmark for internal consistency. To demonstrate the convergent and discriminate validity of our measures, we ran a confirmatory factor analysis (CFA). We also employed a partial least squares (PLS) approach to further strengthen our claim about discriminant validity. The methods and the results are detailed in Appendix B. Both empirical tests provide support for construct validity of our measures.

Dependent Variables

A firm's stock market value reflects the market's expectations of future cash flows to shareholders, and can be viewed as a measure of *perceived* venture performance. This differs from *realized* performance, which is typically embodied in historical measures of firm profitability (e.g., ROI, ROA). Given the level of uncertainty often associated with the true prospects of entrepreneurial firms, perceived performance as stock market value is a particularly suitable measure for an entrepreneurship setting (Stuart et al. 1999). Measures of realized performance such as ROI, ROA, or Tobin's q are less appropriate for young, high-growth entrepreneurial firms that often have negative earnings, few tangible assets, and low (even negative) book values. For instance, 134 firms in our sample (i.e., 86% of the sample firms for which we had the relevant accounting data) had negative earnings in the fourth quarter (Q4) of 1999. Five firms (i.e., 3% of the sample firms for which we had the relevant data) even had a negative book value in the same period. These numbers did not change substantially in Q4 2000.

There are limitations in using stock market valuation as a dependent variable. The nature of our sample and the period in which we collected the data could prompt

concerns about bias, due to an irrational bubble in the stock market. However, while the rationality of the markets during 1999–2000 remains an open question (e.g., Pastor and Veronesi 2004 offer a rational explanation for investors' behavior and provide empirical evidence against the bubble hypothesis), our paper is not predicated on the efficiency of capital markets. It centers on the differential performance implications of alternative business model designs, and our estimation method exploits their differential valuation by capital markets. Even if the companies in our sample were systematically overvalued, this would not distort the qualitative results of our regression analysis. An additional support to our methodology is the fact that our dependent variable captures market participants' perceptions of the business cycle, and the level of resource munificence in the environment. It reflects the factors that market participants value (akin to Shleifer and Vishny's 1991 analysis of stock market valuations of conglomerates in the 1960s and 1980s). This was actually beneficial for our analysis because it allowed us to test our contingency hypotheses about the moderating effect of resource munificence on the relationship between business model design and firm performance.

Because most firms in our sample have relatively low levels of debt, the market value of a firm's equity is a good approximation of the market value of the whole firm. We measured the market value of equity at a given date as the number of shares outstanding multiplied by the firm's stock price, taken from the combined CRSP and Datastream databases. We then calculated the logarithm of the market value of the equity in order to comply with the normality assumption of ordinary least squares (OLS) regression. Following this transformation, we found that the null hypothesis of normality could not be rejected at the 5% level of significance using a Shapiro-Wilk test. To test our hypotheses, we used measurements of the dependent variable at various points in time (annual average, average during Q4, and the last day of trading of Q4) and in various time periods (1999, 2000) characterized by different levels of resource munificence for entrepreneurial firms.

Until now, most empirical research has employed industry-type measures of munificence, such as mean annual industry sales growth (Tushman and Anderson 1986), employment growth in the industry (Dess and Beard 1984), and other indicators of growth at the industry level (McArthur and Nystrom 1991). Because the business model construct spans industry boundaries, and many of the sample firms span multiple industries, we could not define an industry-level variable that captured resource munificence adequately. We therefore measured the dependent variables in time periods that were sufficiently distinct in terms of environmental resource munificence, yet close to the point in time when the independent variables were measured.

Despite the short window, the change in the resource availability for entrepreneurial firms triggered by the worldwide crash of high-tech stocks in March 2000 was severe. Park and Mezias (2005), for example, demonstrate the sharp and statistically significant reversal in a number of munificence measures. Table 1 summarizes the differences between the time periods we considered. The year 1999 (and Q4 1999 in particular) was a time of relatively high munificence for entrepreneurial firms in our sample, whereas the year 2000 (and Q4 2000 in particular) was a time of relatively low munificence. In addition, other indicators of environmental uncertainty, such as complexity and dynamism (Dess and Beard 1984), may have changed between the years, perhaps to a lesser extent.

The use of multiple measures of the dependent variable provided a robustness check for our results. In our analysis, we contrast the average market value of firms in Q4 1999 with that in Q4 2000, the market value of firms at the close of Q4 1999 with that at the close of Q4 2000, and the average market value of firms in 1999 with that in 2000.

Control Variables

We included further factors that might influence the market value of a firm's equity as control variables in the analysis, because their omission might confound it. Our industry controls were the level of competitive threat, and estimated market size. Our raters measured competitive threat on a four-point Likert scale based on information found in annual reports, prospectuses, competitors' SEC documents and websites, Forrester benchmark studies, Hoover's database (which lists each focal firm's main competitors), and investment analysts' reports. The information on market size was obtained from Forrester research reports and from the U.S. Department of Commerce. Consistent with market power arguments (Porter 1980), and our own theory, the greater the level of competition that a business model is facing (in more competitive or smaller markets), the lower the chances that the business model will create much total value and the lower the performance of the focal firm.

Our firm-level controls included the age of the firm, size, country of origin, and expenditures on R&D, advertising, and capital. Size was measured as a logarithm of the number of employees. The variable can be viewed as a proxy for the focal firm's bargaining power, relative to rival firms and other business model stakeholders. All other things being equal, the larger the focal firm, the greater its potential for value creation as well as its bargaining power, and, hence, the better its performance. We controlled for country of origin using a dummy ("1" for firms headquartered in North America, "0" for European firms).

Table 1 Indicators of Resource Munificence 1999 and 2000

Indicators of resource munificence	
1999	2000
• Median quarterly sales growth of sample companies: 30% (Q2 1999), 29% (Q3 1999), 33% (Q4 1999) • Number of Internet-related IPOs in United States: 193 (Q4: 62) • Public market Internet IPO financings in % of total IPO financings: 67% • VC funding for B2C e-commerce companies: \$4.5 billion (+1,000% from 1998) (1)	• Median sales growth of sample companies: 18% (Q1 2000), 15% (Q2 2000), 8% (Q3 2000), 6% (Q4 2000) • Number of Internet-related IPOs in United States: 122 (Q4: 0) (2) • Public market Internet IPO financings in % of total IPO financings: 36% (2) • VC funding for e-commerce companies dropped from \$843 million (Q1) to \$69 million (Q4) (3)

Sources. (1) PriceWaterhouseCoopers, <http://www.ecommercetimes.com/perl/story/2505.html>. (2) Morgan Stanley, 2002. *The Technology IPO Yearbook*, 8th ed. *22 Years of Tech Investing*. http://www.morganstanley.com/institutional/techresearch/tech_ipo_yearbook.html?page=research. (3) PricewaterhouseCoopers/VentureOne, Money Tree Survey Q4 2000.

The inclusion of these firm-level variables strengthens the claim that our analysis captures the influence of distinct business model design characteristics on firm performance, as opposed to the effects of firm characteristics or strategy. For example, investment in R&D has been used in prior research as a proxy for technology strategy (Dowling and McGee 1994) and as a proxy for the degree to which a firm pursues a product differentiation strategy (Mizik and Jacobson 2003). Advertising expenditures have been employed as a proxy for a firm's marketing strategy (Mizik and Jacobson 2003).

Finally, we considered alternative business model design themes, such as complementarities and lock-in (Amit and Zott 2001), by constructing two latent control variables, using 9 indicators for complementarities (Cronbach alpha = 0.70), and 15 indicators for lock-in (Cronbach alpha = 0.74).

Econometric Modeling and Estimation Approach

We analyzed the data using multivariate regression techniques. We tested the robustness and validity of our model specification in several distinct ways. First, we tested for multicollinearity among independent variables by calculating variance inflation factors (VIF) (see Kleinbaum et al. 1998). Second, we performed analyses using different dependent variables. Third, we discarded influential observations based on established criteria for identifying influential points (e.g., leverage, studentized residual, or change in the determinant of the covariance matrix) from our data set to see whether they distorted results. Fourth, we tested for overfitting of the data.² Fifth, we considered the potential bias introduced by sampling on the dependent variable by running a truncated regression model (Maddala 1986).³ Sixth, we tested for homoscedasticity using White's test. Seventh, we tested for potential endogeneity (i.e., the concern that business model design could be a choice variable that is correlated with unobservables that are relegated to the error term) by running a 2SLS regression with

instrumental variables and by using the Hausmann test (see Greene 2003, p. 80ff).⁴ Eighth, we performed multiple partial *F*-tests (Cohen and Cohen 1983) to ensure that adding the design novelty and design efficiency variables, as well as their interaction, improved the fit of the model significantly, compared to a baseline model that contained only control variables.

None of these tests gave rise to concern, yet we observed multicollinearity in those regressions where the interaction term between design novelty and design efficiency was included. We therefore mean-centered the interaction variable, as well as the design novelty and design efficiency measures (see Aiken and West 1991). This significantly reduced the VIF to levels that attenuated the concern about multicollinearity. In addition, we ensured that the mean-centering approach did not entail a lack of invariance of regression coefficients, which may arise in equations containing interactions even under simple linear transformations of the data (Aiken and West 1991). Overall, therefore, we conclude that our model specification proved robust and valid.

Results

Descriptive Statistics

Table 2a, provides an overview of the data set we assembled. It reveals the entrepreneurial nature of our sample firms as well as the enormous change that occurred in the environment between Q4 1999 and Q4 2000. Specifically, in 1999 the median age of a sample company was just over four years, while the mean company was just under seven years old. The few older firms in the sample are those that went through an extensive transformation, with entrepreneurial management leading the change. The median sales of sample companies in 1999 were just under \$25 million, while the median book value of equity in the same year was \$57 million. The median sample company employed 269 people (mean 1,067). With respect to the change in the environment

between 1999 and 2000, we note that the median company was worth \$349 million at the end of December 1999, but only \$49 million at the end of December 2000, representing a decline of 85.6% in market value over a 12-month period.

Table 2b, depicts the Pearson correlations among the right-hand side variables used in the regression analysis. We note that while some correlations among the explanatory variables are significant, they do not pose a multicollinearity problem, as their variance inflation factors (VIF) are low.

Hypotheses Tested

Table 3 depicts the OLS regression results. Tables 3a and 3b (full sample) show the results for regressions in which the dependent variable is the logarithm of market value averaged over Q4 1999 (Table 3a), and Q4 2000 (Table 3b). Table 3c summarizes the main regression results for each of the three dependent variables we considered. Table 3d (restricted sample) depicts the results of the same regressions reported in Tables 3a and 3b on a restricted sample of firms that were present in both 1999 and 2000. In other words, in the regressions reported in Table 3d we control for entry and exit in our sample between 1999 and 2000.

Hypothesis 1 (regarding novelty-centered business model design) is supported by the analysis. As depicted by Table 3 (Tables 3a–3d), the coefficient on the design novelty variable is positive, and in most cases it is significant both during a period of environmental munificence and during a period of resource scarcity. The observed effect was relatively robust to changes in the environment. Our results suggest that even in times of resource

scarcity and less uncertainty about the viability of business model designs, innovative business model designs were associated with higher levels of performance.

Comparing Tables 3a and 3b (full sample) with Table 3d (restricted sample), we note that the coefficient on the design novelty variable is significant at the 1% level in the restricted sample for all four models (see Table 3d), while it is significant at the 10% level in the full sample for three of the four models we ran (see Tables 3a and 3b). This might suggest a weakening of the novelty effect due to entry dynamics. We explored these apparent differences by running a separate set of regressions using only the 30 firms that entered our sample in 2000. (These regressions are available from the authors upon request.) We observed that in most models the coefficients for novelty-centered business model design were not significant for this small set of 30 firms. That is, the novelty-centered business model design of the firms that entered our sample in 2000 did not significantly explain the variance in the dependent variable. Furthermore, according to the *t*-test suggested by Cohen and Cohen (1983, p. 111), the coefficient on design novelty in the sample of the 30 entering firms was significantly different from the respective coefficient in the sample in 1999.

This analysis highlights the potential role of entry dynamics for the hypothesized contingent effect of munificence on the relationship between business model design and firm performance. Specifically, under conditions of low resource munificence, capital markets may be less receptive to new public offerings from firms that center their value proposition on novel business models. Overall, however, Hypothesis 2 (about the changing strength

Table 2a Descriptive Statistics

Variable name (acronym)	Mean	Median	Standard deviation	Min	Max	No. observations
Market value at close of Q4 1999 US\$ million (MVQtr4Close_99)	\$1,506	\$349	\$3,184	\$2	\$25,942	159
Market value at close of Q4 2000 US\$ million (MVQtr4Close_00)	\$387	\$49	\$1,101	\$0.7	\$8,885	173
Design efficiency	0.702	0.712	0.112	0.404	0.92	190
Design novelty	0.366	0.359	0.133	0.077	0.795	190
Complementarity	0.617	0.639	0.174	0.000	0.972	190
Lock-in	0.454	0.463	0.140	0.167	0.763	190
Age of firm	7.0	4.3	7.8	0.4	45.8	190
Ln number of employees	5.723	5.593	1.336	2.833	10.342	190
Country (1 = United States, 0 = European country)	0.88	1.00	0.32	0.00	1.00	190
R&D expense US\$ 00 (million)	\$2.7	\$0.5	\$6.4	\$0.0	\$67.3	190
Advertising expense US\$ 00 (million)	\$4.7	\$1.0	\$9.3	\$0.0	\$52.8	190
Capital expense US\$ 00 (million)	\$42.7	\$3.7	\$415.9	\$0.0	\$5,733.1	190
Book value of equity 99 (million)	\$163.7	\$57.3	\$416.6	−\$68.6	\$4,601.2	188
Book value of equity 00 (million)	\$272.8	\$71.2	\$685.2	−\$967.3	\$5,752.2	160
Sales net US\$ 99 (million)	\$263.3	\$24.9	\$1,575.4	\$0.0	\$20,111.8	177
Sales net US\$ 00 (million)	\$331.7	\$52.9	\$1,643.0	\$0.0	\$20,609	177
Number of employees	1,067	269	3,557	17	31,000	190
Market size US\$ 00 (million)	\$20,477	\$5,400	\$65,640	\$120	\$744,000	190

Table 2b Pearson Correlation

Variable name (acronym)	Centered design efficiency	Centered design novelty	Centered design efficiency and centered design novelty	Complementarities	Lock-in	Competition	Log(market size)	Age	Log(employees)	Country	R&D ex. 00	Adv. ex. 00	Capital ex. 00
Independent variables													
Centered design efficiency	1.000												
Centered design novelty	0.175*	1.000											
Interaction between centered design efficiency and centered design novelty	−0.057	−0.041	1.000										
Control variables													
Complementarity	0.349**	0.349**	−0.001	1.000									
Lock-in	0.316**	0.413**	−0.173*	0.373**	1.000								
Competition	−0.006	−0.322**	−0.147*	−0.128*	−0.121†	1.000							
Ln market size	−0.039	0.016	0.022	0.112	−0.062	0.097	1.000						
Age of firm	−0.112	−0.131†	0.118	0.026*	−0.152*	0.048	0.214**	1.000					
Ln number of employees	0.004	−0.038	0.027	0.040	−0.028	−0.076	0.338**	0.452**	1.000				
Country	−0.074	0.168*	0.123†	0.107	0.159*	0.097	0.462**	0.107	0.110	1.000			
R&D expense 00	0.066	0.220**	0.049	0.088	0.140†	−0.011	0.022	−0.027	0.281**	0.123†	1.000		
Advertising expense 00	−0.088	0.020	0.037	0.042	0.002	0.027	0.199**	0.223**	0.493**	0.178*	0.434**	1.000	
Capital expense 00	0.004	0.018	−0.004	0.147*	0.075	0.055	0.066	0.355**	0.271**	0.036	0.001*	0.414**	1.000

*0.01 ≤ p < 0.05, ** p < 0.01, †0.05 ≤ p < 0.1.

of the design novelty coefficient in different environments) receives little support from our data. Following Gatignon (2003), we examined the moderating role of environmental munificence by conducting a Chow test for the equality of the coefficients in the overall model between the 1999 and 2000 regressions. The test provided significant results (see Table 4). This led us to further examine whether the coefficient on design novelty caused the observed structural break suggested by Table 3c. The table shows that the regression coefficients on the design novelty variable were highly significant in 1999, but less so in 2000. To confirm whether this effect was statistically significant, we conducted a series of pooled regression runs (for all models and all dependent variables): (a) on a completely unrestricted model, in which we included a year dummy (zero for 1999, one for 2000) that we interacted with all variables; (b) on a partially restricted model, in which the only difference from the model in (a) was that the coefficient on novelty-centered business model design was restricted to be the same for 1999 and 2000. Then, we did F -tests to test the null hypothesis of homogeneity of the coefficient on design novelty in models (a) and (b). Following Gatignon (2003, p. 74) the

test statistic we used was $[(PRSS - CUSS)/(DF_{PR} - DF_{CU})]/[CUSS/DF_{CU}]$, where $PRSS$ was the sum of squared residuals from the partially restricted model, $CUSS$ was the sum of squared residuals from the completely unrestricted model, DF_{PR} was the number of degrees of freedom of the partially restricted model, and DF_{CU} was the number of degrees of freedom of the completely unrestricted model. As a result of these tests, we could not reject the null.

Hypothesis 3 (about efficiency-centered business model design) receives mixed support from our data. The results in Table 3 indicate that Hypothesis 3 is supported by the Q4 2000 results in the full sample (Table 3b: Models 1–4). The results are robust across all dependent variables (see Table 3c). In our full sample, during a period of resource scarcity, entrepreneurial firms performed better if their business model design—and hence value proposition to their customers, partners, and suppliers—included efficiency enhancements that reduced their transaction costs, simplified transactions, and sped up processes. However, Hypothesis 3 is not supported by the data pertaining to Q4 1999. Table 3a, which depicts the regression results during a period of environmental munificence (1999), shows that while the

Table 3a Mean-Centered OLS Regression Results (Full Sample)

RHS variables	Dependent variable Ln(market value Q4 av. 1999)			
	Model 1 Value (std. err.)	Model 2 Value (std. err.)	Model 3 Value (std. err.)	Model 4 Value (std. err.)
Constant	19.75***	17.25***	19.78***	17.51***
Design efficiency	1.28 (1.09)	0.93 (0.90)	1.28 (1.09)	1.01 (0.90)
Design novelty	3.32*** (0.93)	2.29** (0.83)	3.21*** (0.93)	2.17* (0.83)
Interaction between design efficiency and design novelty			−8.28 (7.63)	−9.16 (6.09)
Complementarities		−0.63		−0.59
Lock-in		0.80		0.51
Competition		0.05		−0.08
Log(market size)		−0.16*		−0.16*
Age		−0.05***		−0.05***
Log(employees)		0.65***		0.64***
Country		0.12		0.20
R&D expenditure 1999		0.08**		0.09***
Advertising expenditure 1999		0.03		0.03
Capital expenditure 1999		0.00		0.00
R-squared	0.10	0.52	0.11	0.53
Adjusted R-squared	0.09	0.48	0.09	0.49
F	8.47***	13.27***	6.04***	12.53***
N	158	158	158	158

*0.01 ≤ *p* < 0.05, ***p* < 0.01, ****p* < 0.001.**Table 3b Mean-Centered OLS Regression Results (Full Sample)**

RHS variables	Dependent variable Ln(market value Q4 av. 2000)			
	Model 1 Value (std. err.)	Model 2 Value (std. err.)	Model 3 Value (std. err.)	Model 4 Value (std. err.)
Constant	18.42***	16.36***	18.44***	16.64***
Design efficiency	2.21 [†] (1.18)	2.44* (1.01)	2.16 [†] (1.10)	2.51* (1.01)
Design novelty	1.72 [†] (1.01)	1.54 [†] (0.93)	1.70 [†] (1.01)	1.47 (0.93)
Interaction between design efficiency and design novelty			−8.24 (8.45)	−10.25 (6.83)
Complementarities		−0.71		−0.67
Lock-in		−0.30		−0.57
Competition		0.88 [†]		−1.04*
Log(market size)		0.04		0.04
Age		−0.01		−0.01
Log(employees)		0.66***		0.65***
Country		−1.12**		−1.01**
R&D expenditure 2000		0.05**		0.05**
Advertising expenditure 2000		0.02		0.02
Capital expenditure 2000		0.00		0.00
R-squared	0.04	0.47	0.05	0.48
Adjusted R-squared	0.03	0.44	0.03	0.44
F	3.84*	12.59***	2.88*	11.88***
N	180	180	180	180

*0.01 ≤ *p* < 0.05, ***p* < 0.01, ****p* < 0.001, [†]0.05 ≤ *p* < 0.1.

coefficient of the mean-centered design efficiency variable is positive, it is not significant. During this period of abundant resources, efficiency-centered business model design did not contribute to any significant differentiation between entrepreneurial firms.

There is also a lack of support for this hypothesis from the regressions done on the restricted sample

(see Table 3d). The coefficient on design efficiency is insignificant in the restricted sample for each of the models we ran, yet as noted above, it is significant in the full sample for Q4 2000. What accounts for this difference? We probed deeper into the underlying reasons by running a separate set of regressions using only those 30 firms that entered the sample in 2000. We

Table 3c Summary of Main Regression Results for Different Dependent Variables

	Independent variable	Dependent variable	Model 1	Model 2	Model 3	Model 4
1999	Design efficiency	Market value Q4 close	1.70	1.35	1.70	1.44
		Market value Q4 average	1.23	0.93	1.23	1.01
		Market value annual average	1.25	0.89	1.25	0.96
2000	Design efficiency	Market value Q4 close	2.76*	3.16**	2.70*	3.26**
		Market value Q4 average	2.19 [†]	2.44*	2.14 [†]	2.51*
		Market value annual average	1.96*	1.82*	1.88 [†]	1.87*
1999	Design novelty	Market value Q4 close	3.44***	2.61**	3.34***	2.49**
		Market value Q4 average	3.25***	2.29**	3.14***	2.17*
		Market value annual average	2.70**	2.00*	2.62**	1.89*
2000	Design novelty	Market value Q4 close	1.28	1.77 [†]	1.23	1.66
		Market value Q4 average	1.69 [†]	1.54 [†]	1.67 [†]	1.47
		Market value annual average	2.33**	1.93*	2.30**	1.85*

*0.01 ≤ *p* < 0.05, ***p* < 0.01, ****p* < 0.001, [†]0.05 ≤ *p* < 0.1.

Table 3d Mean-Centered OLS Regression Results (Restricted Sample)

RHS variables	Dependent variable Ln(market value Q4 av. 1999)			
	Model 1 Value (std. err.)	Model 2 Value (std. err.)	Model 3 Value (std. err.)	Model 4 Value (std. err.)
Constant	19.79***	17.25***	19.81***	17.50***
Design efficiency	0.89 (1.12)	0.57 (0.93)	0.91 (1.12)	0.68 (0.93)
Design novelty	3.56*** (0.97)	2.51** (0.87)	3.46*** (0.97)	2.40** (0.87)
Design efficiency* design novelty			−6.34 (7.78)	−7.59 (6.18)
Complementarities		−0.35		−0.34
Lock-in		0.37		0.13
Competition		−0.03		−0.15
Log(market size)		−0.18*		−0.18*
Age		−0.05***		−0.05**
Log(employees)		0.69***		0.68***
Country		0.21		0.28
R&D expenditure 1999		0.08**		0.08**
Advertising expenditure 1999		0.02		0.02
Capital expenditure 1999		0.00		0.00
<i>R</i> -squared	0.11	0.54	0.11	0.55
Adjusted <i>R</i> -squared	0.09	0.5	0.09	0.5
<i>F</i>	8.25***	12.70***	5.71***	11.89***
<i>N</i>	142	142	142	142
RHS variables	Dependent variable Ln(market value Q4 av. 2000)			
	Model 1 Value (std. err.)	Model 2 Value (std. err.)	Model 3 Value (std. err.)	Model 4 Value (std. err.)
Constant	18.42***	15.95***	18.43***	16.16***
Design efficiency	1.36 (1.32)	1.57 (1.09)	1.37 (1.32)	1.67 (1.09)
Design novelty	2.97** (1.13)	2.83** (1.02)	2.93* (1.13)	2.74** (1.02)
Design efficiency* design novelty			−2.39 (9.13)	−6.8 (7.17)
Complementarities		0.37		0.38
Lock-in		−2.04 [†]		−2.25*
Competition		−0.51		−0.61
Log(market size)		−0.03		−0.04
Age		−0.01		−0.01
Log(employees)		0.69***		0.68***
Country		−0.43		−0.36
R&D expenditure 2000		0.06**		0.06**
Advertising expenditure 2000		0.02		0.02
Capital expenditure 2000		0.00		0.00
<i>R</i> -squared	0.07	0.52	0.07	0.53
Adjusted <i>R</i> -squared	0.05	0.48	0.05	0.48
<i>F</i>	4.88**	11.74***	3.25**	10.90***
<i>N</i>	142	142	142	142

*0.01 ≤ *p* < 0.05, ***p* < 0.01, ****p* < 0.001, [†]0.05 ≤ *p* < 0.1.

Table 4 Chow Test for Structural Break Between 1999 and 2000

Dependent variable	Model 1	Model 2	Model 3	Model 4
Market value Q4 close	34.1**	14.92**	25.69**	14.06**
Market value Q4 average	18.3**	9.1**	13.81**	8.6**
Market value annual average	2.48 [†]	1.93*	1.95	1.87*

Notes. Table entries are *F*-statistics.

* $0.01 \leq p < 0.05$, ** $p < 0.01$, [†] $0.05 \leq p < 0.1$.

observed that these firms had significant positive coefficients for efficiency-centered business model design in 2000, which were also significantly different from the respective coefficients in the restricted sample in 1999 (according to the *t*-test suggested by Cohen and Cohen 1983, p. 111). This may have strengthened the design efficiency effect in the full sample.

This analysis highlights the role of entry dynamics. Under low resource munificence, capital markets may be more receptive to new public offerings from firms that promise lower transaction costs. This in turn may favor the IPOs of firms that have more efficiency-centered business model designs. Overall, however, Hypothesis 4 has little support from our empirical analysis. While the Chow test suggests a structural break in the overall model parameters between 1999 and 2000 (see Table 4), and while Table 3c suggests a strengthening of the design efficiency effect from 1999 to 2000, a series of pooled regression analyses with dummy variables (see also Gatignon 2003, p. 74) did not allow us to reject the null hypothesis that the coefficient on design efficiency was identical in 1999 and 2000.⁵

Hypothesis 5 (about the interaction effect among the design themes of business models) received no support from the empirical analysis: None of our regressions revealed a significant positive interaction. Indeed, the coefficient of the variable capturing the interaction between design novelty and design efficiency had a negative sign in all the regressions we ran, yet in most cases it was not statistically significant (see Tables 3a, 3b, and 3d). That coefficient, however, was significant at the 10% level in Model 4 for the full sample when we used the logarithm of the market value of firms at the close of Q4 2000, and the logarithm of the average market value of firms in 2000, as dependent variables. For these models we performed post hoc analysis using plotting techniques suggested by Aiken and West (1991). The plots of design efficiency on the respective dependent variable for different values of design novelty revealed that for higher values of design novelty, the slope of the plotted regression line was smaller, but remained positive. In other words, the plot was consistent with Hypothesis 6 and yielded the additional insight that while diseconomies of scope in design might exist, they do not override the positive effects of efficiency-centered design on performance. Our

analysis of novelty-centered design yielded analogous results. Our data seem to suggest, yet do not convincingly prove, that there appear to be diseconomies of scope in design; that is, attempting to emphasize both efficiency and novelty in the design of a business model may be costly and could adversely affect performance. Our analysis provides preliminary, albeit statistically weak, support for Hypothesis 6.

Discussion and Conclusion

The central thesis anchoring our study is the notion that organizational design should extend beyond internal design (Nystrom and Starbuck 1981) to include a focus on the architecture of the transactions that a focal firm engineers with its partners, suppliers, and customers. Consistent with Foss (2002), we suggest the need to pay greater attention to the structuring of firm boundaries and, in particular, to the structuring of a firm's exchanges with external stakeholders. We develop a theory of business model design that explains how value is created at the business model level of analysis and how it is captured at the focal firm level of analysis. No prior theory explicitly centered on this issue exists. Our theoretical contribution is the model that links business model design to performance of entrepreneurial firms under varying conditions of environmental munificence.

Although the design of the business model has been raised as an important issue for research on new organizational forms and boundary-spanning organizational designs (Foss 2002, Rindova and Kotha 2001), it has not been explored in detail. Perhaps this is because, until now, we lacked a methodology for conceptualizing and measuring business model design with a high degree of granularity. Our methodological contribution is that we provide a way to think about, and measure, business model design themes. By moving beyond generic typologies of business models (which are often only applicable to e-commerce firms), we offer a greater level of abstraction and a higher degree of granularity in the description and measurement of business model designs. This allows us to outline the design elements of business models that are relevant for wealth creation, and it also allows us to provide measures that can be more generally applied to entrepreneurial firms.

Our research is particularly relevant to the study of new organizational forms, innovation, and entrepreneurship. It adds to an emerging body of research on business models (e.g., Amit and Zott 2001, Chesbrough and Rosenbloom 2002, Hargadorn and Douglas 2001, Mendelson 2000). We operationalize and measure the business model construct, and show empirically that it has an impact on wealth creation. To the best of our knowledge, this is the first rigorous empirical large-sample study of business model design themes. While there are promising empirical studies in this domain

(e.g., Rajgopal et al. 2003), there has been no systematic large-scale empirical analysis of the performance implications of business model design themes under various environmental regimes.

By providing common definitions for new organizational forms (particularly boundary-spanning designs), our study suggests design options and enables comparisons among different designs, enhancing the scientific base of organization design (Dunbar and Starbuck 2006). It can also help bridge the growing gap between the reality of organization design and organization theory that some scholars have identified (e.g., Daft and Lewin 1993, Ilinitich et al. 1996). While organization scholars have investigated intriguing, yet largely isolated, cases of new organizational forms, such as the dynamic network (Miles and Snow 1986), the virtual corporation (Davidow and Malone 1992), or hypertext organizations (Nonaka and Takeuchi 1995), research on these designs could be unified and advanced through common frameworks, concepts, and theories (Daft and Lewin 1993, Foss 2002). Our study shows that capturing configuration as variables helps us characterize and measure business model designs, which should facilitate further research in that area.

This paper articulates essential features and properties of boundary-spanning organizational designs, and addresses their performance implications. Our strongest and most robust finding relates to the novelty-centered business model design. This centers on innovation, which is “the specific instrument of entrepreneurship. It is the act that endows resources with a new capacity to create wealth” (Drucker 1985, p. 30). Wealth-creating innovation may be achieved through a recombination of existing resources (Schumpeter 1934) in new designs. Our study shows that firms can innovate not only by recombining the resources they control, but also by harnessing those of the partners, suppliers, and customers who participate in their business model. In this way, our study contributes to the entrepreneurship literature. We highlight business model design as a crucial task for entrepreneurs, and as a source of innovation. We also find that environmental munificence does not moderate the positive relationship between business model design innovation and focal firm performance. This counterintuitive finding is noteworthy. It attests to the remarkable temporal stability of that relationship, emphasizing the business model as an important and enduring locus of innovation and wealth creation.

By framing business model design as an entrepreneurial task, and by identifying business model innovation as a source of wealth creation for firms, our work informs research at the intersection of organization theory, entrepreneurship, and strategy (Hitt et al. 2001, Langlois 2005). Business model-specific effects may account for some hitherto unexplained variance in the performance of firms. In this sense, they complement,

but do not replace, firm-specific and industry-specific effects on firm performance (Rumelt 1991, Hawawini et al. 2003, McGahan and Porter 2002). We corroborate the premise that in a highly interconnected world, entrepreneurs should consider looking beyond firm and industry boundaries in order to create and capture business opportunities. They can create wealth by introducing innovative boundary-spanning organization designs.

We acknowledge several limitations of this study. Some empirical results could be affected by measurement problems. For example, our measurement of business model design themes may not have captured all the lines of a firm’s business that have revenue potential, and therefore might not explain all the variation in the dependent variable due to business model design themes. Another problem could be that bad management corrupts inherently good designs. Unfortunately, our data do not allow us to control for the quality of management. Data limitations also prevent us from engaging in a dynamic analysis of business model evolution or measuring value creation directly at business model level. Last, the scope of the theory presented in this paper, and the data set used to test it, do not allow us to draw generalizable conclusions about the role of business model designs in the broader population of firms.

From an entrepreneurial standpoint, however, the limitations of this study could present interesting opportunities for future research. For example, do our results apply to more mature and established organizations? What factors give rise to and shape business model designs? How do regulations, customer preferences, and competition influence the emergence and evolution of these designs? What are the dynamics of business model design change, and how stable are business model designs across time? How reliable is the impact of various business model design themes on performance, and do efficiency-centered business models have higher reliability of performance than novelty-centered ones (Sorensen 2002, Sorenson and Sorensen 2001)? Organization scholars might be particularly interested in how the firm’s architecture of boundary-spanning transactions is linked to its internal organization and how the interaction of the two affects firm performance. Strategy scholars could be interested in whether business model design contributes to the competitive advantage of firms, and how it interacts with firm strategies, such as product market positioning.

We hope that the ideas presented in this paper will inspire and enable further research on these intriguing issues. We believe that the perspective of the business model, its design elements, and the concepts developed here for describing and measuring business model design themes are a step toward an improved understanding of boundary-spanning organizational designs.

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Appendix B. Convergent and Discriminant Validity

We first ran a confirmatory factor analysis (CFA) on a measurement model with two factors, where the design efficiency traits loaded onto the design efficiency factor, and the design novelty traits loaded onto the design novelty factor. In this model, the correlation between the design efficiency and the design novelty index was estimated. We then ran a CFA on a measurement model with only one factor, where the correlation between the design efficiency and the design novelty variable was constrained to be one. If the model where the correlation is not equal to one improves the fit significantly compared to the constrained model, the two constructs (i.e., design novelty and design efficiency) are distinct from each other, although they can be significantly correlated (Gatignon et al. 2002, Gatignon 2003).

We also used CFA to establish the convergent validity of the constructs, by comparing a measurement model where the correlation between the two constructs was estimated with a model where the correlation was constrained to be equal to

Appendix A. Survey Items

Survey item	Scale*
Efficiency-centered business model design	
Inventory costs for participants in the business model are reduced.	SA, A, D, SD
Transactions are simple from the user's point of view.	SA, A, D, SD
The business model enables a low number of errors in the execution of transactions.	SA, A, D, SD
Costs other than those already mentioned for participants in the business model are reduced (i.e., marketing and sales costs, transaction-processing costs, communication costs, etc.).	SA, A, D, SD
The business model is scalable (i.e., can handle small as well as large number of transactions).	SA, A, D, SD
The business model enables participants to make informed decisions.	SA, A, D, SD
Transactions are transparent: Flows and use of information, services, goods can be verified.	SA, A, D, SD
As part of transactions, information is provided to participants to reduce asymmetric degree of knowledge amongst them regarding the quality and nature of the goods being exchanged.	SA, A, D, SD
As part of transactions, information is provided to participants about each other.	SA, A, D, SD
Access to large range of products, services, information, and other participants is provided.	SA, A, D, SD
The business model enables demand aggregation.	Y, N
The business model enables fast transactions.	SA, A, D, SD
The business model, overall, offers high transaction efficiency.	SA, A, D, SD
Novelty-centered business model design	
The business model offers new combinations of products, services, and information.	SA, A, D, SD
The business model brings together new participants.	SA, A, D, SD
Incentives offered to participants in transactions are novel.	SA, A, D, SD
The business model gives access to an unprecedented variety and number of participants and/or goods.	SA, A, D, SD
The business model links participants to transactions in novel ways.	SA, A, D, SD
The richness (i.e., quality and depth) of some of the links between participants is novel.	SA, A, D, SD
Number of patents that the focal firm has been awarded for aspects of its business model.	0, 1–2, 3–4, >4
Extent to which the business model relies on trade secrets and/or copyrights.	R, S, B, N
Does the focal firm claim to be a pioneer with its business model?	Y, N
The focal firm has continuously introduced innovations in its business model.	SA, A, D, SD
There are competing business models with the potential to leapfrog the firm's business model.	SA, A, D, SD
There are other important aspects of the business model that make it novel.	SA, A, D, SD
Overall, the company's business model is novel.	SA, A, D, SD

*SA—Strongly Agree (coded as 1); A—Agree (0.75); D—Disagree (0.25); SD—Strongly Disagree (0); Y—Yes (1), N—No (0); R—Radically (1); S—Substantially (0.66); B—a bit (0.33), N—not at all (0); 0 (0), 1–2 (0.33), 3–4 (0.66), >4 (1).

Table B.1 Confirmatory Factor Analysis

Correlation	Chi-squared	Degrees of freedom
0.14	733.4	319
0	735.1	320
1	761.9	320

zero. “A significant improvement in fit indicates that the two constructs are indeed related, which confirms convergent validity” (Gatignon et al. 2002, p. 1109).

We used LISREL to implement the CFA, following the routines described in Gatignon (2003, pp. 178–220). The results from the CFA are displayed in Table B.1.

The results from the CFA demonstrate that efficiency-centered design and novelty-centered design are two distinct dimensions of business models, although they are positively correlated (estimated correlation = 0.14). This is confirmed by a significantly (at the 0.01 level) improved confirmatory factor analytic model when the correlation is estimated, compared to a measurement model where the correlation is constrained to 1 (chi-squared = $761.9 - 733.4 = 28.5$, degrees of freedom = $320 - 319 = 1$). Furthermore, the results from the CFA demonstrate that efficiency-centered design and novelty-centered design are independent dimensions of business models. The confirmatory factor analytic model when the correlation is estimated, compared to a measurement model where the correlation is constrained to zero, is not significantly improved (chi-squared = $735.1 - 733.4 = 1.7$, degrees of freedom = $320 - 319 = 1$). This is akin to the Gatignon et al. (2002) result that some dimensions of innovation (e.g., competence-enhancing/destroying) are independent of others (e.g., radicalness), yet all measure important, distinct aspects of innovation.

In addition to CFA, the literature suggests partial least squares (PLS) as another method for assessing discriminant validity. Using PLS, one can determine whether a construct shares more variance with its measures than it shares with other constructs in the model (Hulland 1999, Reinartz et al. 2004). This is achieved (1) by calculating the square roots of the average variance extracted (AVE) values, which measure the average variance shared between a construct and its measures, and (2) by calculating the correlations between different constructs. A matrix can then be constructed where the square root of AVE is in the diagonal, and the correlations between the constructs are in the off-diagonal. For adequate discriminant validity, the diagonal elements should be greater than the off-diagonal elements in the corresponding rows and columns (Fornell and Larcker 1981). In our case, we obtained

Table B.2 Partial Least Squares Analysis

	Design efficiency	Design novelty
Design efficiency	0, 233	0, 175
Design novelty	0, 175	0, 243

Notes. (1) The square root of AVE is displayed in the diagonal, and the correlations between the constructs are displayed in the off-diagonal. (2) According to Fornell and Larcker (1981), if the factor variance is set to one, then the average variance extracted is defined as: $AVE = \sum \lambda_i^2 / (\sum \lambda_i^2 + \sum (1 - \lambda_i^2))$.

the matrix depicted in Table B.2 as a result of the PLS analysis. The results further strengthen the discriminant validity of our constructs.

Endnotes

¹There are many studies that examine a range of issues relating to the conduct and performance of e-commerce firms that became publicly listed corporations during the late 1990s (see, for example, Rajgopal et al. 2002, 2003; Kotha et al. 2004). Put together, these and related studies provide important insights into the type of firm we investigate in this paper. Data limitations prevented us from using Heckman’s technique to correct for potential survival bias.

²Overfitting occurs when the fit of the model with the data is due to the idiosyncrasy of a specific data set. We took a random subsample of 150 firms, with the remaining firms constituting the holdout sample, then calibrated the model based on the subsample and applied the resulting parameter estimates to the holdout sample, calculating goodness-of-fit, pseudo F -value, and Theil U statistic.

³To use the truncated regression model (see Maddala 1986), we assume that private entrepreneurial firms have smaller market values than public ones. We find that the coefficient estimates from the truncated regression are analogous to the OLS estimates. The Wald test in the truncated regression, and the F -statistic in the OLS regression, are both significant at the 5% level for all models.

⁴An appendix that describes the method is available on request from the authors.

⁵These results also fail to support the idea that capital markets valued novelty- versus efficiency-centered business model designs differentially before and after the stock market crash on NASDAQ in April 2000 due to irrational assumptions. We would like to thank one reviewer for this insight.

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